# JIANGZI CHEN

(480)-415-9946 Jchen722@asu.edu

## Education

**Nanjing University of Information Science and Technology (NUIST)** 9/2021-6/2025
Bachelor of Science in Physical Geography and Resource Environment;
GPA: 3.46/4 (83)

**Arizona State University (ASU)** 8/2025-Present
Master of Science in Civil, Environmental and Sustainable Engineering;
GPA: 3.5/4.0

## Publications

Wan, Zhaoyi, Chengyi Zhao, Jianting Zhu, Xiaofei Ma, **Jiangzi Chen**, and Junhao Wang. 2024. "Assessment and Prediction of Coastal Ecological Resilience Based on the Pressure–State–Response (PSR) Model" *Land* 13, no. 12: 2130. https://doi.org/10.3390/land13122130

Impact of Spartina Alterniflora Expansion on the Tidal Creek Formation in the Mudflat of Yellow Sea (submitted)

## Project Experiences

**Unpermitted On-site Wastewater Treatment Facility (OWTF) Identification for Rural Arizona Environmental Health** 1/2026-Present

*Supervisor: Ruijie Zeng, ASU*

- Gis-based spatial analysis on Arizona stateside Wastewater Treatment Facility (OWTF) distribution
- A novel classification and identification method for identifying potential OWTF.

**Calculation of Ecological Water Use in the Tarim River Basin** 6/2024-7/2025

*Supervisor: Yaojie Liu, NUIST*

- Extracted NDVI in Tarim River Basin by GEE
- Removed outliers by interpolation and other methods and conducted data cleaning and preparation for further analysis

**Assessment and Prediction of Coastal Ecological Resilience Based on the Pressure-State-Response (PSR) Model** 2/2024-7/2025

*Supervisor: Prof. Chengyi Zhao, NUIST*

- Developed a new approach based on the Pressure-State-Response (PSR) model to assess and predict pixel-scale multi-year ecological resilience (ER)
- Applied this approach to investigate spatiotemporal variations of ER in the China's coastal zone (CCZ) in the last few decades and predict future ER trend under various SSP-RCP scenarios

**Impact of Spartina Alterniflora Expansion on the Tidal Creek Formation in the Mudflat of Yellow Sea** 10/2023-8/2024

*Supervisor: Prof. Guanghui Zheng, NUIST*

- Utilized high-resolution remote sensing data and remote sensing technology to study the impact of Spartina Alterniflora on the evolution of tidal creek
- Conducted dynamic monitoring of the invasion of Spartina Alterniflora in the Yancheng Red-crowned Crane Reserve
- Studied a series of ecological problems arising from the invasion of Spartina Alterniflora, and revealed the relationship between different variables, such as the distribution and density of Spartina Alterniflora, and the characteristic parameters, such as the bifurcation ratio (Rb) , curvature and density of the tidal creek

**Multiple Decision-making Regional Resilience Analysis based on MADM Model from the Perspective of Disasters** 9/2022-7/2024

*Supervisor: Associate Prof. Jiaxun Li, NUIST*

- Collected data through extracting basic geographical information data of Jiangning District, Nanjing, using ArcGIS and ENVI
- Analyzed the data through the MADM model (a hybrid multi-decision analysis model)
- Compared regional resilience and identified differences in assessment results that affect regional resilience, proposed suggestions on improving village resilience and reducing losses in disasters based on the research results

**Research on the Evolution and Optimization of Coastal Zone Ecological Pattern based on Cellular Automata,** 3/2023-3/2024

*Leader, Supervisor: Prof. Chengyi Zhao, NUIST*

- Proposed an integrated system coupling the PLUS model and InVEST model
- Predicted the pattern of land use changes and quantitatively assessed the habitat quality from 2020 to 2050 under three SSP-RCP scenarios (SSP126, SSP245 and SSP585) in China's coastal zone
- Evaluated the driving forces of changes of habitat quality and contribution of land cover changes to the degradation of habitat quality

## Teaching Experience

**Teaching Assistant - CEE 441** Spring 2026

*Arizona State University*

- Held regular office hours to answer students' questions.
- Graded assignments.

## Professional Experiences

**Monitoring and Evaluation of the Rural Planning Initiative in Jiangsu Province, GTMAP** 7/2024-8/2024

*Implementation Engineer Intern*

- Identified and checked satellite maps of site types, calculated area statistics for each site type, and determined each type's potential development areas
- Conducted initial neighborhood planning, including the positioning of enterprises, farmlands, transportation lines, and residential areas, as well as the interplay between the relationship between industries, and the ease of living in order to set up a rational layout

**Nanjing Bureau of Planning and Natural Resources** 1/2023-2/2023

- Conducted information sorting and field inspection of geological disaster sites in Gulou District, Nanjing, and inspected 22 geological disaster sites in Gulou District
- Evaluated the existing problems, potential hazards and safety risks to the surrounding residents of geological disaster sites, and classified the danger level of the geological disaster sites based on their situations
- Monitored in real-time of site types within the Gulou District for the illegal uses or the wrong purposes of land use
- Verified information on floor space, plans, etc. for new residential areas

## Extracurricular Activities

***Member*, Baseball and Softball Team, NUIST** 9/2021-9/2022

- Participated in school-level and provincial-level competitions

***Member*, Student Union, Changwang School of Honors, NUIST** 9/2021-7/2022

- Dealt with external public relations affairs and promoted activities of the student union, discovered and utilized off-campus resources to provide students with more learning and development opportunities
- Coordinated internal and external communications to ensure smooth and accurate information and progress of large-scale events

## Skills

**Software:** ArcGIS, ENVI, MATLAB, Fragstats, PLUS, InVEST, Python